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Argamal: Malware hidden in hentai games

Malware Analysis

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Argamal: Malware hidden in hentai games

1. Executive Summary

Argamal is a Windows remote access Trojan (RAT) delivery chain identified by Kaspersky in June 2026 after researchers observed trojanized adult-game downloads distributing a previously undocumented implant. The infection path hides inside otherwise functional game archives, abuses modified runtime libraries, establishes delayed persistence, then retrieves and decrypts a second-stage RAT that provides broad remote control over the victim host.

The campaign matters because it combines social distribution through niche gaming sites and torrents with a comparatively durable post-compromise chain: PowerShell staging, scheduled task execution, COM hijacking persistence, GitHub-hosted payload retrieval, and fallback C2 changes tied to locale checks. The resulting malware supports system control, file theft, screenshots, command execution, and uploads, making a confirmed detection both a malware event and a probable identity/data exposure incident.

Defenders should prioritize telemetry that links user-launched game content to modified DLL loading, PowerShell execution, bitsadmin-driven downloads from GitHub, suspicious CLSID and InprocServer32 changes under HKCU, and unusual outbound traffic over UDP 57441 or TCP 3747. Incident response should assume the attacker may have obtained credentials, screenshots, local files, and durable remote access.

2. Sample Metadata

Field	Value
Family	Argamal
Type	Windows RAT / downloader chain
Platform	Windows
First publicly reported	Kaspersky Securelist, 03 Jun 2026
Initial lure theme	trojanized "hentai" game downloads
Distribution channels observed	niche game websites, PixelDrain-hosted archives, and torrent trackers including AniRena
Confidence	medium overall because the technical chain is well documented by vendor research, but public reporting is still concentrated around a single primary analytic source
Modified FFmpeg DLL SHA1	42add9475e67a1ccc6a6af94b5475d3defc01b85
Stage loader DLL SHA1	edce72f59e4c1d136cd1946af70d334c19df858d
Additional downloader SHA1	9803604ec45f31f9ef75bcca1e1310d8ac1fc3a6
Infrastructure pivot	186[.]158[.]223[.]35 behind multiple observed C2 domains

Operational handling summary:

- Treat confirmed execution as likely full host compromise, not a nuisance adware event.
- Expect credential theft, remote command execution, screenshot collection, and file staging.
- Preserve registry, scheduled-task, PowerShell, BITS, and user-profile file-write telemetry early in triage.

3. Source Review & Confidence

Source	Contribution	Confidence
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Kaspersky Securelist technical write-up by Mikhail Reznichenko for the infection chain, malware capabilities, infrastructure, and hashes.	Primary analytical contribution	High / Medium by analyst review
Kaspersky press release for high-level campaign scope, victim geography, and operator assessment context.	Primary analytical contribution	High / Medium by analyst review
MITRE ATT&CK technique references used to normalize observed behaviors into defender-friendly categories.	Primary analytical contribution	High / Medium by analyst review
Secondary media coverage used only for corroborative framing, not for unique technical claims.	Primary analytical contribution	High / Medium by analyst review

Confidence assessment:

What should carry most decision weight:

- High confidence that Argamal uses trojanized game archives, modified DLL loading, PowerShell staging, delayed second-stage execution, and COM hijacking persistence because these behaviors are described consistently in the primary Kaspersky research.
- Medium confidence in the broader scale and attribution assessment because public victim counts are approximate and the Spanish-speaking actor hypothesis is explicitly assessed only with medium confidence.
- Medium confidence in infrastructure durability because the reporting already shows domain and payload changes during the investigation period.
- The chain from user-launched game archive to staged PowerShell and COM hijack persistence.
- The host-compromise severity implied by full RAT functionality.
- The likelihood that static blocklists alone will age quickly as delivery paths and C2 values change.

4. Payload Variants and Delivery

Observed delivery paths center on archives containing legitimate game content plus one maliciously modified library. The initial Securelist example used a trojanized game archive that bundled a patched `ffmpeg.dll`. Because the game legitimately needs FFmpeg at runtime, the malicious DLL loads automatically when the victim starts the game, preserving expected gameplay and reducing suspicion.

Kaspersky also observed alternate packaging patterns:

- a payload embedded as `libpython64.dat` or similarly named data files in `lib\py3-windows-x86\_64`
- modified game components used to load that embedded payload
- a malicious downloader DLL posted on a gaming forum while disguised as a cheat

Delivery implications:

- The actor can pivot between websites, torrents, and forum posts without changing the post-exploitation goal.
- The modified runtime-library approach is likely to outperform simple filename-based blocking because the surrounding archive still contains legitimate game files.
- Download-source reputation, user-execution telemetry, and unexpected DLL behavior are higher-value controls than any single hash.

5. Infection Chain Analysis

Representative infection chain:

1. A victim downloads a trojanized game archive from a game site, PixelDrain-backed download, torrent tracker, or forum post.
2. Launching the game causes the modified `ffmpeg.dll` to load.
3. The DLL calls `DllGetClassObject` from `natives2\_blob.bin`, which acts as a DLL-based script executor.
4. `natives2\_blob.bin` runs a Base64-encoded PowerShell script (Stage1).
5. Stage1 checks for basic analysis artifacts, seeds persistence-related registry values and environment variables, and creates a scheduled task that runs three days later.
6. Stage2 retrieves encrypted payload `zaesdl.dat` from GitHub via `bitsadmin.exe`, decrypts it with AES-CBC, and writes the resulting DLL into a random subdirectory under `%USER%\AppData\Local`.
7. Stage2 registers the decrypted DLL under `HKCU\SOFTWARE\Classes\CLSID\{B210D694-C8DF-490D-9576-9E20CDBC20BD}\InprocServer32`, hijacking the Windows Color System Calibration Loader COM object so the malware executes at user logon.
8. The final payload beacons to the C2, can receive commands, and can escalate into an extended RAT mode over TCP.

Why this matters:

- The chain intentionally delays second-stage execution, which can split early user execution from later network activity.
- Persistence is tied to a legitimate Windows scheduled task and COM registration path, both of which deserve focused hunting.
- GitHub-hosted encrypted payload retrieval can blend into normal enterprise traffic if teams only look for obviously malicious domains.

6. Technical Analysis

Key observed behaviors:

- DLL side-loading via a modified game dependency
- PowerShell-based staging with Base64-encoded scripts
- Environment-variable and registry-backed handoff between stages
- Delayed scheduled-task execution
- AES-CBC decryption of a GitHub-hosted payload
- COM hijacking persistence at user logon
- Broad RAT functionality for system control, discovery, file operations, screenshots, and uploads

6.1 Loader and staging details

The Stage1 PowerShell script checks for controlled environments including Sandboxie folders and Procmon64 in the process list. If the host does not appear instrumented, Stage1 stores another Base64-encoded script in the `MI\_V` environment variable (and `MI\_V2` in newer variants), writes random DLL paths under a user-specific `AppData\Local` subdirectory, and creates a scheduled task that triggers three days later.

Stage2 uses those registry-stored paths to retrieve encrypted content `zaesdl.dat` from GitHub using `bitsadmin.exe`. The payload is decrypted with AES-CBC using key `zbcd1j9234r670eh` and the same value as the IV, then written to disk as the DLL that later backs COM hijack persistence.

6.2 Persistence and execution model

The downloader temporarily uses CLSID `{722D0F89-B69C-4700-AE8C-4A44350E4876}` during staging, then cleans up that path after Stage2 runs. The lasting persistence mechanism instead targets CLSID `{B210D694-C8DF-490D-9576-9E20CDBC20BD}`, a COM object used by the Windows Color System Calibration Loader scheduled task. That design gives the malware logon-triggered execution without requiring overt startup-folder artifacts.

6.3 RAT capability summary

The payload supports:

- system reboot and shutdown control

- command execution via ShellExecuteW, CreateProcessW, and task-based execution
- screenshot capture via `SCREEN` / `SCREEN9`
- directory creation, file rename, deletion, and archival (`ZIPFILE`, `ZIPFOLDER`, `TAR`, `TAR2`)
- file upload to alternate servers and file download or collection behavior
- system, drive, username, and idle-time reconnaissance
- mouse and keyboard interaction commands in extended mode

The malware also checks for dozens of security products through `tasklist` output and adapts some infrastructure behavior based on locale, including a documented `zh-CN` branch that switched the C2 domain to `country1[.ignorelist[.com]`.

7. Indicators of Compromise

High-value campaign indicators captured in `iocs.csv` include: Operational guidance:

Type	Value	Description
sha1	42add9475e67a1ccc6a6af94b5475d3defc01b85	Modified ffmpeg.dll loader observed in trojanized game archive
sha1	edce72f59e4c1d136cd1946af70d334c19df858d	natives2_blob.bin DLL script executor
sha1	9803604ec45f31f9ef75bcca1e1310d8ac1fc3a6	Trojan downloader sample hash
sha1	02819d200d1424882af81cb504b3e8614b32397a	Additional Trojan downloader sample hash
sha1	76253fb55aed707440e808ea78e7101318436b1c	Argamal RAT payload sample hash
sha1	1405a3c5e0aeb08012484134e16cdec4ab29b4a4	Argamal RAT payload sample hash
sha1	535f4337f261b6da20a3c614eb13270bed2d533a	Argamal RAT payload sample hash
sha1	d2cb0d7a9ad2b5d4ea7c2da8aec62beb37cf36d6	Argamal RAT payload sample hash

8. MITRE ATT&CK Mapping

Mapped ATT&CK techniques from the supporting artifact:

- T1204.002 - User Execution: Malicious File
- T1574.002 - Hijack Execution Flow: DLL Side-Loading
- T1059.001 - Command and Scripting Interpreter: PowerShell
- T1497.001 - Virtualization/Sandbox Evasion: System Checks
- T1053.005 - Scheduled Task/Job: Scheduled Task
- T1197 - BITS Jobs
- T1105 - Ingress Tool Transfer
- T1112 - Modify Registry
- T1546.015 - Event Triggered Execution: Component Object Model Hijacking
- T1082 - System Information Discovery
- T1083 - File and Directory Discovery
- T1113 - Screen Capture
- T1095 - Non-Application Layer Protocol

Interpretation:

- The chain combines user execution, staged scripting, delayed execution, and persistence techniques that are individually common but collectively distinctive when tied to niche game lures.
- Defenders should correlate the early-stage loader behaviors with the later COM hijack and C2 traffic rather than alerting on each behavior independently.

9. Detection Engineering

Detection priorities: Supporting sidecar artifacts: Recommended analytic themes:

Signal	Telemetry Source	Analytic Goal
Unsigned or suspicious `ffmpeg.dll` loads from user-downloaded game folders followed by child PowerShell execution.	EDR / process / registry / network telemetry	Identify delivery, persistence, or C2 evidence
PowerShell spawning from game-related parent processes, especially when Base64 content, environment-variable staging, or CLSID writes are present.	EDR / process / registry / network telemetry	Identify delivery, persistence, or C2 evidence
`bitsadmin.exe` retrieving content from GitHub shortly before new DLL files appear in random `%AppData%\Local` subdirectories.	EDR / process / registry / network telemetry	Identify delivery, persistence, or C2 evidence
Registry writes to `HKCU\SOFTWARE\Classes\CLSID\{722D0F89-B69C-4700-AE8C-4A44350E4876}` or `HKCU\SOFTWARE\Classes\CLSID\{B210D694-C8DF-490D-9576-9E20CDBC20BD}`.	EDR / process / registry / network telemetry	Identify delivery, persistence, or C2 evidence
Outbound UDP 57441 or TCP 3747 traffic from unexpected user-context processes.	EDR / process / registry / network telemetry	Identify delivery, persistence, or C2 evidence

9.1 Embedded Hunt Query

```
// Argamal delivery-chain hunt: suspicious game execution followed by PowerShell or BITS activity
DeviceProcessEvents
| where Timestamp > ago(30d)
| where InitiatingProcessFolderPath has_any ("Downloads", "Desktop", "Temp")
| where FileName in~ ("powershell.exe", "bitsadmin.exe", "rundll32.exe") or ProcessCommandLine has_any ("ffmpeg.dll", "natives2_blob.bin", "zaesdl.dat")
| project Timestamp, DeviceName, InitiatingProcessFileName, FileName, ProcessCommandLine, FolderPath
| order by Timestamp desc
```

10. Threat Hunting

Hunt objectives:

- Find suspicious game archives or extracted folders that contain runtime libraries not expected for the original title.
- Trace PowerShell, bitsadmin, and registry activity back to user-launched game executables or torrent-derived content.
- Identify machines that reached the observed C2 infrastructure or wrote random DLL paths beneath `%AppData%\Local` shortly before logon persistence events.
- Search for GitHub-hosted payload retrieval followed by DLL load or COM-registration changes.

Useful hunt pivots:

- `ffmpeg.dll` or `natives2\_blob.bin` in unusual game-install or user-download directories
- scheduled tasks created for delayed execution around the initial user launch window
- `MI\_V` or `MI\_V2` environment variable creation in suspicious contexts
- CLSID modifications under HKCU for the documented COM objects
- network telemetry showing outbound UDP 57441 or TCP 3747 from non-standard client software

10.1 Embedded Hunt Query

```
// Argamal persistence and network hunt pivots
DeviceRegistryEvents
| where Timestamp > ago(30d)
| where RegistryKey has_any ("SOFTWARE\Classes\CLSID\{722D0F89-B69C-4700-AE8C-4A44350E4876}",
"SOFTWARE\Classes\CLSID\{B210D694-C8DF-490D-9576-9E20CDBC20BD}")
| project Timestamp, DeviceName, InitiatingProcessFileName, RegistryKey, RegistryValueName,
RegistryValueData
| order by Timestamp desc
```

11. Triage and Incident Response

Immediate response steps:

9. Isolate affected hosts and preserve EDR, registry, scheduled-task, and PowerShell telemetry.
10. Collect the game archive, extracted directory, modified DLLs, scheduled-task artifacts, and any `AppData\Local` random subdirectories referenced by the malware.
11. Reset passwords and invalidate sessions for accounts used on the affected host, with priority on browsers, email, gaming, and enterprise application access.
12. Review outbound connections to the documented domains, IP, GitHub repositories, and any file-upload destinations reached by the process tree.
13. Reimage or surgically remediate only after confirming the COM hijack registry paths and delayed task artifacts have been removed.

Exposure review should explicitly include:

- browser-stored credentials and cookies
- screenshots or locally stored sensitive files that may have been exfiltrated
- commands executed remotely after initial beaconing
- any files compressed or staged for upload by the RAT

12. Validation and Emulation

Use controlled detections and lab emulation for the observable chain elements rather than live malware execution in production networks. High-signal validation targets include simulated HKCU CLSID writes, AppData DLL creation, `bitsadmin.exe` retrieval from GitHub, and parent-child process chains where user-facing game content launches PowerShell.

Safe validation checks:

- confirm your telemetry captures HKCU `InprocServer32` changes for the documented CLSIDs
- confirm `bitsadmin.exe` network events are linked to process lineage and destination domain
- confirm PowerShell command-line logging preserves Base64-heavy staging commands
- confirm hunts can join suspicious DLL writes in user paths to subsequent logon-triggered activity

13. Recommendations

Priority	Owner	Action
Critical	SOC / IR	Add hunts and alerts for the documented COM hijack CLSIDs, unexpected `bitsadmin.exe` GitHub retrievals, and

		user-context traffic over UDP 57441 or TCP 3747.
High	Endpoint Engineering	Increase visibility or controls for unsigned DLL loads from user-download and extracted-game paths, especially where a game binary or launcher immediately triggers PowerShell.
Critical	SOC / IR	Treat Argamal detections as likely credential and session exposure; reset passwords, revoke tokens, and review browser-stored secrets.
Medium	Security Team	Reinforce that unofficial game downloads, mods, cheats, and torrent-hosted installers are common malware delivery vectors and should not be run on business-managed assets.
Medium	Security Team	Build detections around the chain, not only the final RAT payload.
Medium	Security Team	Preserve GitHub-download telemetry even when the hosting domain itself is broadly legitimate.
Critical	SOC / IR	Add game-folder and archive-execution context to threat hunts where consumer software on enterprise systems is unusual.

14. References and Refresh Notes

- Securelist technical research: Argamal: Malware hidden in hentai games
- <https://securelist.com/argamal-rat-distributed-with-hentai-games/119999/>
- Kaspersky press release: Kaspersky discovers Argamal: a new malware hidden in games for adults
- <https://www.kaspersky.com/about/press-releases/kaspersky-discovers-argamal-a-new-malware-hidden-in-games-for-adults>
- MITRE ATT&CK T1546.015 - Component Object Model Hijacking
- <https://attack.mitre.org/techniques/T1546/015/>
- MITRE ATT&CK T1574.002 - DLL Side-Loading
- <https://attack.mitre.org/techniques/T1574/002/>

Intelligence gaps:

- Public reporting does not yet provide complete payload families, compile cadence, or sample chronology across all observed variants.
- Victimology is only described at country level and may understate total impact.
- The medium-confidence attribution assessment remains tentative and should not drive executive conclusions without additional corroboration.